

LESSON 13.3

ANGLE RELATIONSHIPS IN TRIANGLES



LESSON INTRODUCTION

MA.8.GR.1.5 Solve problems involving the relationships of interior and exterior angles of a triangle.

LEARNING TARGETS

- I can identify the relationships of interior and exterior angles of a triangle.
- I can write and solve algebraic equations to solve mathematical problems involving angle relationships of triangles.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.1.3 Prove relationships and theorems about triangles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of triangles.

ESSENTIAL QUESTIONS

- What is the relationship between interior and exterior angles of a triangle?
- How can I use angle relationships to solve problems involving triangles?

LESSON NARRATIVE

This lesson introduces students to the angle relationships that exist in triangles. Students explore interior angles of a triangle, discovering the triangle sum theorem. Students will then engage in a collaborative activity where they work with exterior angles and non-adjacent interior angles.

COMPONENT SUMMARY

13.3.1 WARM-UP (~5 MIN)

Notice and wonder about a miter saw

13.3.3 GUIDED PRACTICE (10 MIN)

Find interior angle measures of a triangle

13.3.5 GUIDED PRACTICE (5 MIN)

Find exterior angle measures of a triangle

13.3.2 EXPLORATION (15 MIN)

Explore the interior angles of a triangle

13.3.4 COLLABORATION (10 MIN)

Collaborate on an activity about exterior angles of triangles

13.3.6 WRAP-UP (~5 MIN)

Self-reflection on the learning targets

RESOURCES

GLOSSARY

Key Terms:

- Triangle Sum Theorem
- Exterior angle

Additional Terms:

- N/A

MATERIALS

- Rectangular paper (i.e., index cards)
- Straightedge

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may mistake the value of the sum of the interior angles of a triangle.
- Sometimes students will believe in solving for x (or an unknown value) that they have found the angle measure. Students should be reminded to pay attention to what they are solving when they are finding an unknown value.

THINGS TO CONSIDER

- 13.3.2 Exploration is an important activity for students to see how the interior angles of a triangle add up to 180° . Go through this activity on your own to identify potential areas where students may struggle.

LITERACY AND LANGUAGE ROUTINES

Notice and Wonder

13.3.1 Warm-Up provides students with an opportunity to discuss what they notice and wonder about a miter saw. Steer the conversation to focus on angles the saw could cut.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

13.3.2 Exploration provides students with a hands-on activity that helps them understand why the sum of the interior angles of a triangle is 180° . Students use a rectangle paper to discover the triangle sum theorem.

DIFFERENTIATE INSTRUCTION

Consider the use of additional digital manipulatives when presenting the information in this lesson. For example, for students who are having difficulty with 13.3.2 Exploration, consider using a digital software such as Desmos, GeoGebra, etc. to help students visualize why the interior angles always add up to be 180° .

13.3.1 WARM-UP: MATH TALKS

TEACHER MOVES

This activity leverages a Notice and Wonder protocol. The purpose of this protocol is to make content accessible for all students with these low-stakes questions. By thinking about them and responding, students gain entry into the context and might get their curiosity piqued.



13.3.1 Warm-Up: Math Talks

1. Angle measures are often used in construction projects. Special saws, like the miter saw shown, are used to make precise cuts into wood and other raw materials.

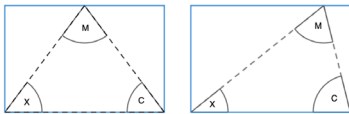


- a. What is something you notice about the miter saw?
Answers will vary. I notice there is a circular ruler at the bottom.
- b. What is something you wonder about the miter saw?
Answers will vary. I wonder how you can line up the miter saw.



13.3.2 Exploration: Interior Angles of a Triangle

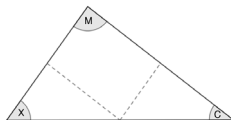
- Complete the following using a piece of rectangular paper.
 - Use a straightedge to draw diagonals from the bottom two corners to intersect on the top side of the paper. Two possible examples are shown.



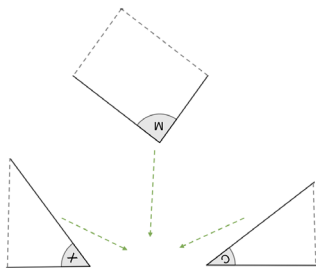
- Label the angle formed where the lines you drew intersect M . Then, label the other angles in the triangle X and C , as shown.

These angles are called the **interior angles** of the triangle because they are *inside* the triangle.

- Cut out the triangle.
- Cut or tear the triangle so that each angle is separated. An example of where to cut is shown.



- Rotate and slide the labeled angles to form a set of three adjacent angles. The image shows the beginning of this process with arrows indicating where the angles should continue to slide.



- What do you notice?
Answers will vary. I notice the three angles form a straight angle. I notice the three angles fit together on a straight line.
- Compare your findings with your partner.
Answers will vary.

- Complete the conjecture about the relationship of the three interior angles of a triangle.
Adjacent angles that form a straight line have a sum of

- ☐ 90° .
- ☒ 180° .
- ☐ 360° .

Based on the findings from the Exploration,

the three interior angles of a triangle sum to

- ☐ 90° .
- ☒ 180° .
- ☐ 360° .

13.3.2 EXPLORATION: INTERIOR ANGLES OF A TRIANGLE

TEACHER MOVES

Materials needed for this activity:

- A piece of rectangular paper or an index card.
- Straightedge

TEACHER MOVES

Consider modeling for students how to create and cut the triangle for questions 1a-1e if you notice students struggling to follow the directions. If doing this, make sure that students create a different triangle from your own.

TEACHER MOVES

When this activity is complete, engage students in a discussion about what they've completed in this activity and their conjecture in question 2. This discussion can help springboard students into the formal definitions of this concept that will be introduced in the next activity.

13.3.3 GUIDED PRACTICE: INTERIOR ANGLES OF TRIANGLES

DIFFERENTIATE INSTRUCTION

Consider creating an anchor chart that provides the concepts of this lesson in a way that students can refer to throughout the remainder of the unit.

ENRICHMENT

How can the triangle sum theorem be used to help solve question 1?

TEACHER MOVES

Pay attention to student performance on questions 2b and 2c. When students are just asked to find the measure of a specific angle, they often will provide the value of x , but not for the angle. Help draw student attention to what the question is asking.

ENRICHMENT

- How can you verify that the angle measure found is correct?
- What should the sum of the measures be?



13.3.3 Guided Practice: Interior Angles of Triangles

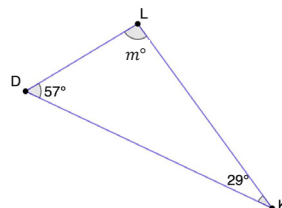
The relationship of the interior angles of a triangle can be used to solve for unknown angle measures in a triangle.



Triangle Sum Theorem

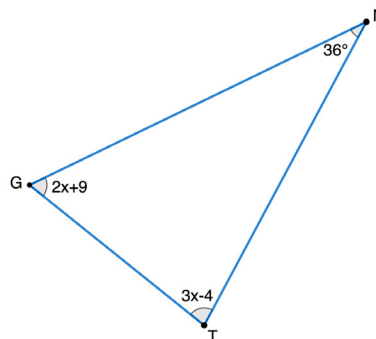
The sum of the measures of the interior angles of a triangle is 180° .

1. Triangle LDK is shown.



- Write an equation that can be used to find the measure of the unknown angle in the triangle.
 $57 + m + 29 = 180$
- Solve the equation to find the measure of $\angle KLD$.
 $m = 94$. The measure of $\angle KLD$ is 94° .

2. Triangle NGT is shown.



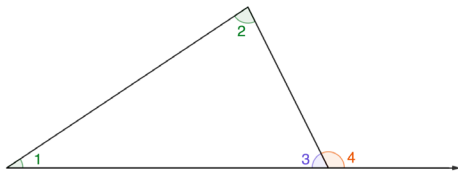
- Write an equation that can be used to find the value of x .
 $2x + 9 + 3x - 4 + 36 = 180$
- Solve the equation.
 $x = 27.8$
- Find the measure of $\angle GTN$.
 79.4° .



13.3.4 Collaboration: Exterior Angles of Triangles

An **exterior angle** is an angle created by extending one leg of a triangle.

In the triangle shown, $\angle 4$ is an **exterior angle** and $\angle 3$ is its **adjacent interior angle**.



Work with your partner to complete the following based on the given information.

- Complete the statements.

An exterior angle is ☐ complementary ☒ **supplementary** ☐ vertical to its adjacent

interior angle, so the sum of $\angle 3$ and $\angle 4$ is ☐ 90° . ☒ **180°** . ☐ 360° .

Angles 1, 2, and 3 are the ☐ **interior** ☐ exterior angles of the

triangle. Their sum is ☐ 90° . ☒ **180°** . ☐ 360° .

The notation $m\angle 1$ is read as "The measure of angle 1."

- Destiny wrote the following mathematical statements on her paper after looking at the triangle above.

$$m\angle 4 + m\angle 3 = 180^\circ$$

$$m\angle 1 + m\angle 2 + m\angle 3 = 180^\circ$$

$$m\angle 1 + m\angle 2 = m\angle 4$$

- With your partner, discuss Destiny's reasoning.
- Complete the statement.

Any exterior angle of a triangle is equal to

- ☐ one of the interior angles of the triangle.
- ☒ **the sum of the two non-adjacent interior angles of the triangle.**

13.3.4 COLLABORATION: EXTERIOR ANGLES OF TRIANGLES

DIFFERENTIATE INSTRUCTION

Add the term "exterior angle" to any Anchor Chart related to this concept of an existing word wall, or similar device in the room.

TEACHER MOVES

Refer students back to their Frayer Model, foldable, anchor chart, or other note-taking that students did from Lesson 1 if they forget the terms complementary, supplementary, and vertical.

TEACHER MOVES

Allow students to collaborate and reason with the notation in question 2. Do not explain the statements for them before they have had a chance to make sense of it for themselves.

ENRICHMENT

- How is Destiny able to state that the sum of the measures of angles 1 and 2 is equal to the measure of angle 4?
- Why do you think that any exterior angle of a triangle is equal to the sum of the two non-adjacent interior angles of a triangle?

13.3.4 COLLABORATION: EXTERIOR ANGLES OF TRIANGLES (CONTINUED)

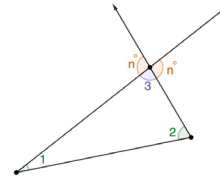
QUESTIONING

Consider having students create possible exterior angles for angles 1 and 2 as well so that students can see how the exterior angles for angle 3 can be created.

TEACHER MOVES

At the end of the activity, engage students in the discussion of their answers in question 3b. The language used here will be very important to connect prior learning to this concept.

3. Every interior angle of a triangle has two possible exterior angles, depending on the side length that is extended. Consider the triangle shown with **both possible exterior angles** of $\angle 3$ labeled n° .



- a. Discuss with your partner the angle relationship between both possible exterior angles of $\angle 3$.
Answers will vary.
- b. Work with your partner to complete the statements.

The two possible exterior angles of $\angle 3$ are

- ☐ complementary
- ☐ supplementary
- ☒ vertical

angles, so they both have the

same measure. Therefore, it does not matter which

side length is extended to create an exterior angle

because both are equivalent to each other and

- ☐ complementary
- ☒ supplementary
- ☐ vertical

to the adjacent interior angle.



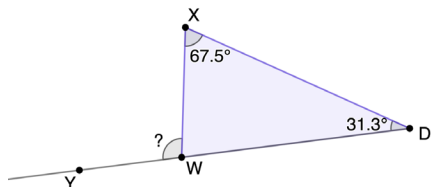
13.3.5 Guided Practice: Exterior Angles of Triangles

The relationship of the exterior angles of a triangle can be used to solve for unknown angle measures in a triangle.



The measure of an exterior angle of a triangle is equal to the sum of the measures of the two non-adjacent interior angles of the triangle.

1. Consider the diagram shown.



Find the measure of $\angle YWX$.

98.8°.

13.3.5 GUIDED PRACTICE: EXTERIOR ANGLES OF TRIANGLES

TEACHER MOVES

The formal theorem name is not included here in this activity. In the state benchmark clarifications for this standard, it is not listed by name. It is important for students to understand how it works. Their knowledge of the formal name is not called for by this standard.

ENRICHMENT

How can you use your knowledge from 13.3.4 Collaboration to solve question 1?

13.3.5 GUIDED PRACTICE: EXTERIOR ANGLES OF TRIANGLES (CONTINUED)

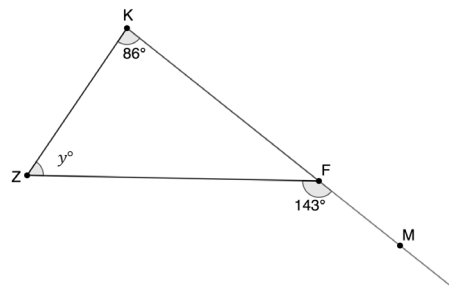
QUESTIONING

For an additional challenge, ask students to determine the measure of angle KFZ.

TEACHER MOVES

After this activity, allow students to share out their answers and to provide a summary of what they've learned in this lesson about interior angles and exterior angles of triangles.

2. A figure is shown.



- Write an equation that can be used to find the value of y .
 $143 = 86 + y$
- Solve the equation.
 $y = 57$



13.3.6 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each learning target listed in the table.



I need help.



I got it.



I could help someone.

Answers will vary.

I can identify the relationship between interior and exterior angles of a triangle.	  
I can write and solve algebraic equations to solve mathematical problems involving angle relationships of triangles.	  

13.3.6 WRAP-UP: REFLECTION

QUESTIONING

What would reflection look like for you as a teacher? Consider your own scale for how you think students responded to your facilitation of the content. Do you think overall students got it or could help someone? If not, what ideas do you have to improve the instruction and teacher-facilitation of the lesson?